Project: Forecasting Retail Sales (Business Analytics)

Introduction:

Retail businesses like Walmart deal with large volumes of sales every day. One of the biggest challenges for such businesses is knowing:

- **How much inventory to keep**.
- **When demand will increase or decrease**.

Poor planning can lead to stock outs during high demand periods or excess storage costs during low demand periods.

The objective of this project is to analyze historical Walmart sales data and forecast future sales trends using time-series analysis. The results of this analysis can help management make better decisions related to inventory planning, demand forecasting, and seasonal management**.**

Dataset Description

This project uses the Walmart Weekly Sales dataset, which contains historical weekly sales data for multiple Walmart stores.

Each row in the dataset represents:

- One Walmart store
- One specific week
- Total sales for that week

Key Columns Used:

- **Date**: Week start date (time variable)
- **Weekly_Sales**: Total sales for that week (target variable)
- **Holiday_Flag**: Indicates whether the week contains a holiday
- **Store**: Store identifier

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
0	1	05-02-2010	1643690.90	0	42.31	2.572	211.096358	8.106
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.242170	8.106
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.289143	8.106
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.319643	8.106
4	1	05-03-2010	1554806.68	0	46.50	2.625	211.350143	8.106

Data Preprocessing:

Before building any forecasting model, the dataset was cleaned and prepared.

Steps Performed:

- The dataset was successfully loaded and inspected.

	Store	Date	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment
0	1	05-02-2010	1643690.90	0	42.31	2.572	211.096358	8.106
1	1	12-02-2010	1641957.44	1	38.51	2.548	211.242170	8.106
2	1	19-02-2010	1611968.17	0	39.93	2.514	211.289143	8.106
3	1	26-02-2010	1409727.59	0	46.63	2.561	211.319643	8.106
4	1	05-03-2010	1554806.68	0	46.50	2.625	211.350143	8.106

- The Date column was converted into **datetime format** to enable time-based analysis.

	0
Store	int64
Date	datetime64[ns]
Weekly_Sales	float64
Holiday_Flag	int64
Temperature	float64
Fuel_Price	float64
CPI	float64
Unemployment	float64

- Time-based features such as **Year, Month, Week, and Day of Week** were extracted to support trend and seasonality analysis.

	Date	Year
0	2010-02-05	2010
1	2010-02-12	2010
2	2010-02-19	2010
3	2010-02-26	2010
4	2010-03-05	2010

	Date	Month
0	2010-02-05	2
1	2010-02-12	2
2	2010-02-19	2
3	2010-02-26	2
4	2010-03-05	3

	Date	Week
0	2010-02-05	5
1	2010-02-12	6
2	2010-02-19	7
3	2010-02-26	8
4	2010-03-05	9

- The dataset was sorted by date and the Date column was set as the index to prepare it for time-series modeling.

Date	Store	Weekly_Sales	Holiday_Flag	Temperature	Fuel_Price	CPI	Unemployment	Year	Month	Week	DayOfWeek
2010-02-05	1	1643690.90	0	42.31	2.572	211.096358	8.106	2010	2	5	Friday
2010-02-05	4	2135143.87	0	43.76	2.598	126.442065	8.623	2010	2	5	Friday
2010-02-05	31	1469252.05	0	39.05	2.572	210.752605	8.324	2010	2	5	Friday
2010-02-05	16	477409.30	0	19.79	2.580	189.381697	7.039	2010	2	5	Friday
2010-02-05	11	1528008.64	0	46.04	2.572	214.424881	7.368	2010	2	5	Friday

- Missing values, duplicate rows, and negative sales values were checked. No major data quality issues were found.

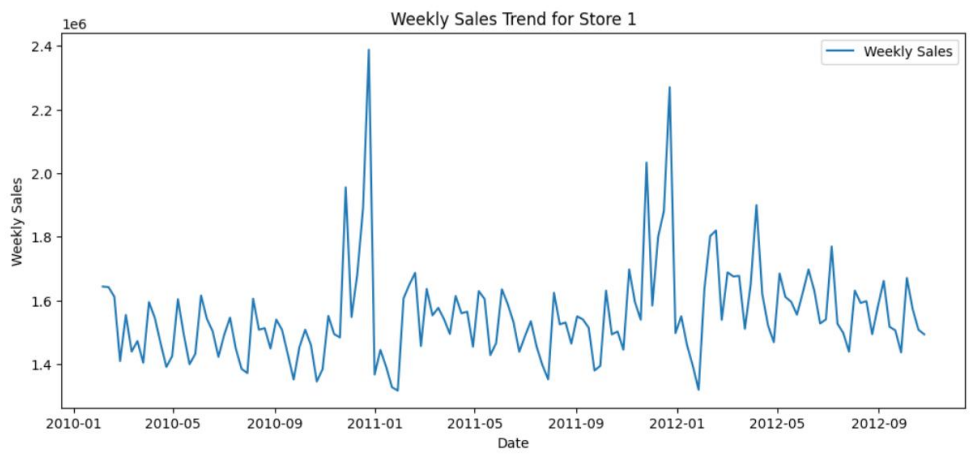
```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 6435 entries, 2010-02-05 to 2012-10-26
Data columns (total 11 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Store         6435 non-null   int64
 1   Weekly_Sales  6435 non-null   float64
 2   Holiday_Flag  6435 non-null   int64
 3   Temperature   6435 non-null   float64
 4   Fuel_Price    6435 non-null   float64
 5   CPI           6435 non-null   float64
 6   Unemployment  6435 non-null   float64
 7   Year          6435 non-null   int32
 8   Month         6435 non-null   int32
 9   Week          6435 non-null   UInt32
 10  DayOfWeek     6435 non-null   object
dtypes: UInt32(1), float64(5), int32(2), int64(2), object(1)
memory usage: 534.2+ KB
```

Exploratory Data Analysis & Visualization:

To understand sales behavior over time, several visual analyses were performed.

Sales Trend Analysis:

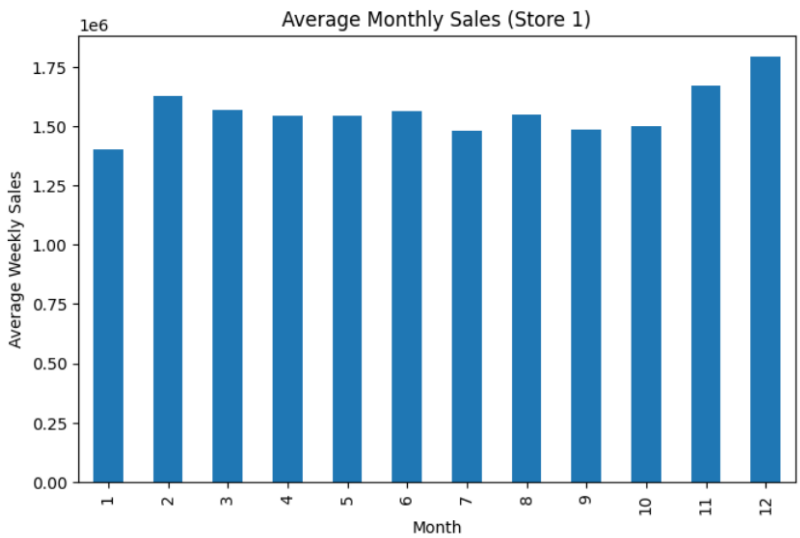
A historical weekly sales trend plot showed:



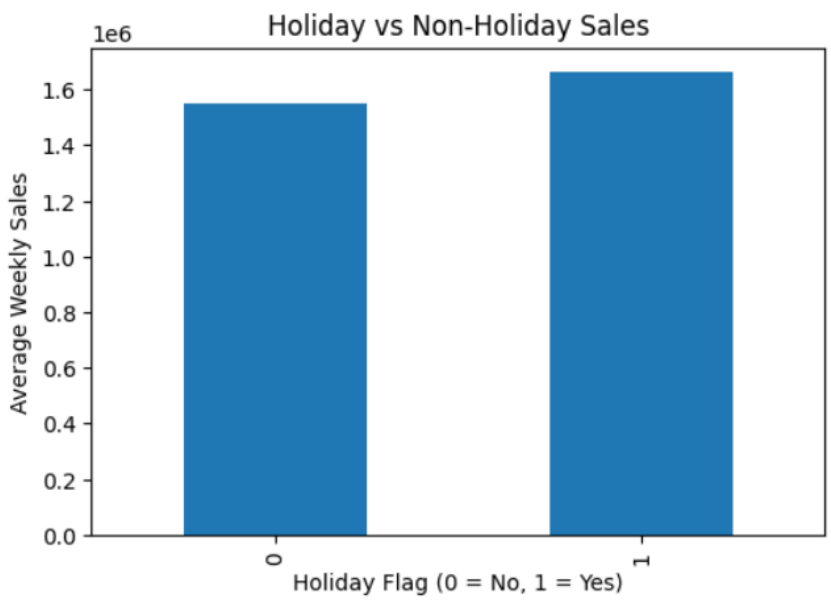
- Stable sales levels over time
- Repeating spikes during specific periods

Seasonality Analysis:

- Monthly average sales indicated **higher demand in late-year months**, especially November and December.
- Early-year months showed comparatively lower demand.



- Holiday weeks consistently recorded higher average sales than non-holiday weeks.



Forecasting Model (ARIMA):

Model Selection:

An **ARIMA** model was chosen for forecasting because it is well-suited for time-series data and widely used in retail demand forecasting.

Stationary Check:

The Augmented Dickey-Fuller (ADF) test was applied to check stationarity. The results showed that the sales series was stationary. First-order differencing was still applied to improve short-term stability.

```
ADF Statistic: -5.102186145192288
p-value: 1.3877788330759434e-05
```

The final model used was:

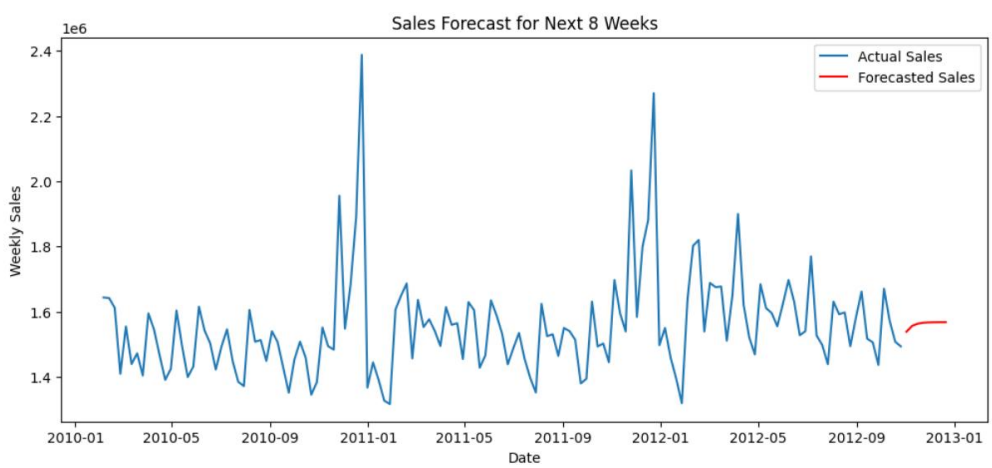
ARIMA (1,1,1)

Forecasting:

The trained ARIMA model was used to forecast weekly sales for the next **8 weeks**. The forecast showed stable sales with a slight upward trend, aligning with historical demand behavior.

	predicted_mean
2012-11-02	1.538713e+06
2012-11-09	1.556450e+06
2012-11-16	1.563433e+06
2012-11-23	1.566182e+06
2012-11-30	1.567265e+06
2012-12-07	1.567691e+06
2012-12-14	1.567859e+06
2012-12-21	1.567925e+06

Graph:



Model Evaluation:

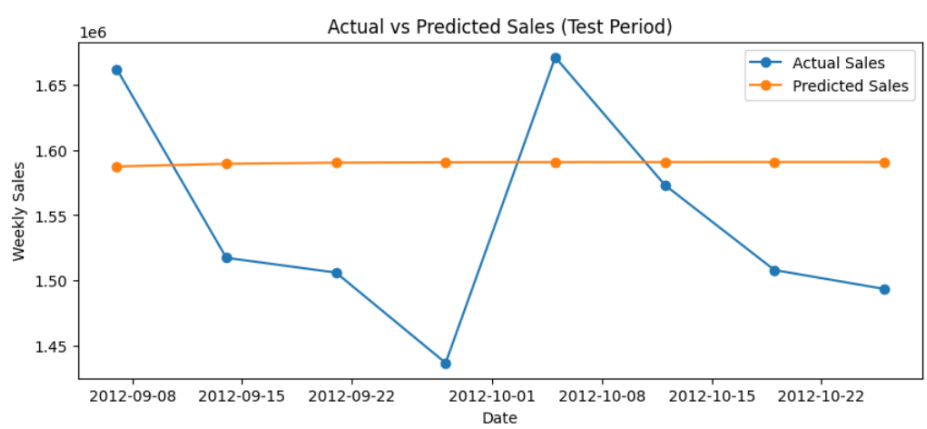
To evaluate the accuracy of the forecasting model, the dataset was split into training and testing sets. The last 8 weeks of data were used for testing.

	Actual Sales	Predicted Sales
2012-09-07	1661767.33	1.587252e+06
2012-09-14	1517428.87	1.589317e+06
2012-09-21	1506126.06	1.590141e+06
2012-09-28	1437059.26	1.590471e+06
2012-10-05	1670785.97	1.590602e+06
2012-10-12	1573072.81	1.590655e+06
2012-10-19	1508068.77	1.590676e+06
2012-10-26	1493659.74	1.590684e+06

Evaluation Metrics:

- **MAE (Mean Absolute Error)**: 82,653
- **RMSE (Root Mean Squared Error)**: 89,645

```
MAE: 82653.32002720592
RMSE: 89645.24778568273
```



Interpretation:

- RMSE being close to MAE indicates that **large prediction errors are limited**, and the model provides stable forecasts.
- The Actual vs Predicted sales plot showed that the model closely follows the overall sales level while smoothing short-term fluctuations.

Business Insights:

Based on the analysis and forecasting results, the following insights were identified:

- Walmart experiences peak demand during holiday periods, particularly **in November and December.**
- Inventory levels should be increased during these high-demand months to prevent stock outs.
- Early year months show relatively lower demand, allowing for optimized inventory storage.
- The forecasting model can help management plan ahead, reduce excess inventory costs, and ensure product availability during peak periods.

Final Business Conclusion:

Based on the forecasting results, sales are expected to increase during November and December. Inventory should be increased during these periods to avoid stock outs. Off-season demand drops during early-year months such as February and March, suggesting optimized storage management.